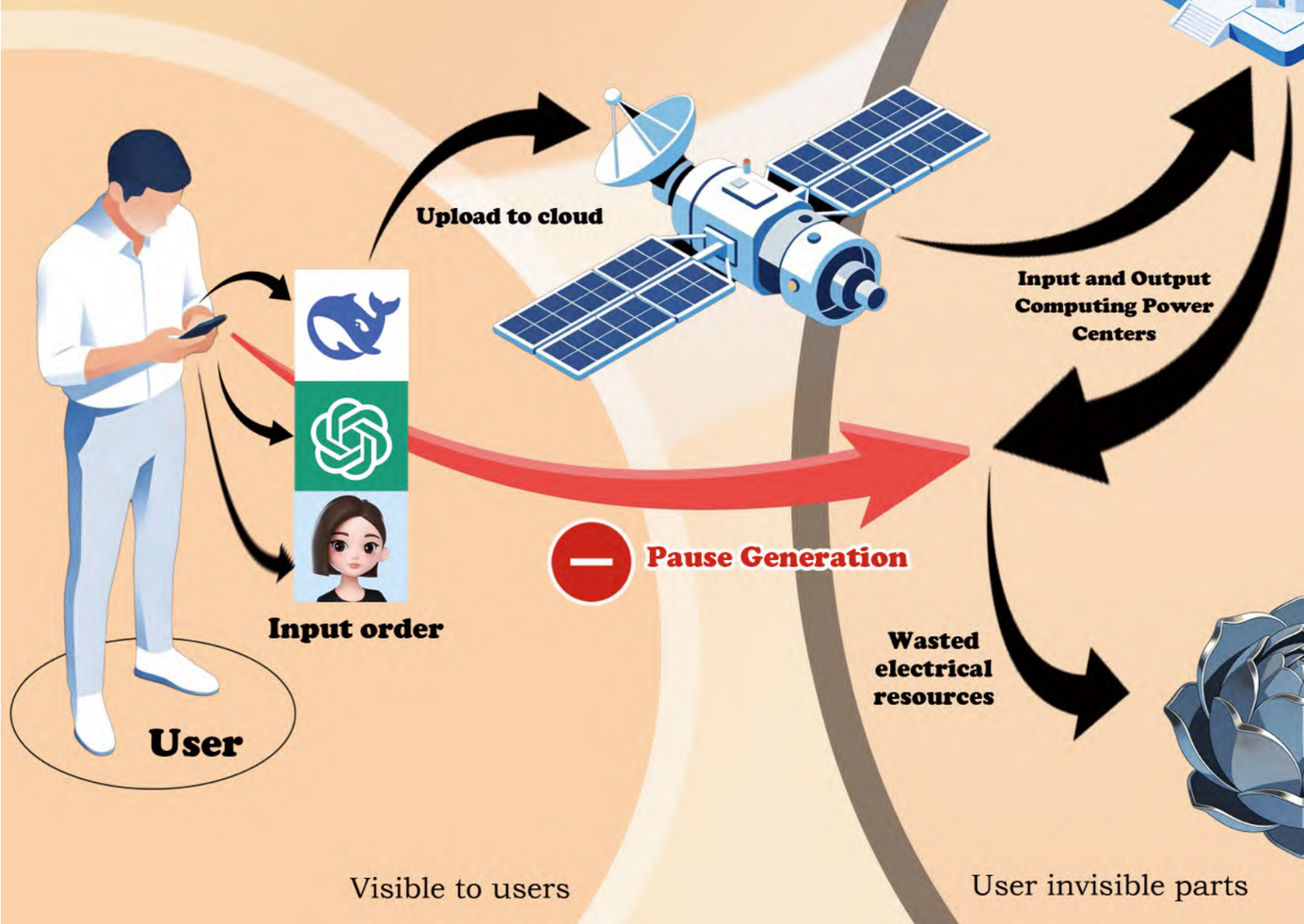
The Shape of Computing Power

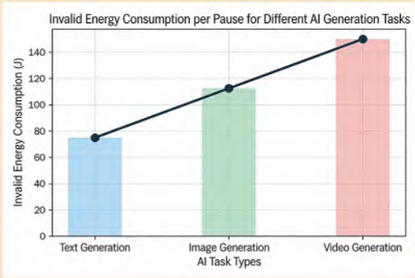
Pause Behavior-driven Energy Consumption Visualization Interaction Design

During AI text, image, and video generation, users often abort the process early due to unsatisfactory results, misoperations, or the need for revisions. Statistics show that the interruption rate is 15%–25% for text, 25%–40% for images, and up to 30%–50% for videos. On average, one out of every 3–5 generation tasks is interrupted.

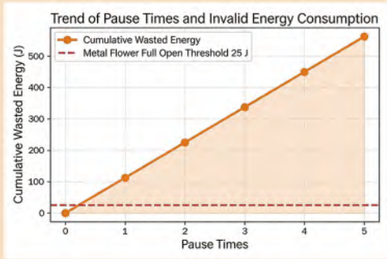
As the pause command is a **mandatory instantaneous instruction**, these interrupted requests do not cease immediately. Instead, they continue to occupy computing resources until the system completes background processing. Aborting an image generation still consumes about 60% of the full computing power, while video interruption leaves up to 45% redundant computation.



Supply electricity



As AI-generated tasks grow in complexity (from text to images to videos), GPU power consumption increases, resulting in a significant rise in ineffective energy consumption per pause.



The memory flower requires only 25 J of energy to fully bloom. If the user pauses just once, the wasted energy already exceeds the threshold, causing the flower to bloom immediately. As the number of pauses increases, the ineffective energy consumption rises linearly, significantly worsening energy waste.

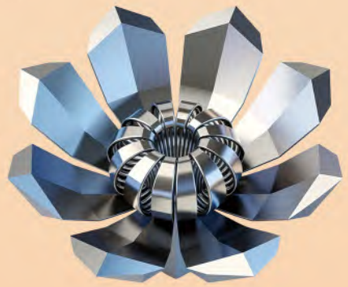
During AI generation, the GPU remains under sustained high computational load. When users pause operations, the server cannot stop immediately and continues to maintain elevated power consumption for a short period, resulting in unnecessary energy waste. Take common devices as examples:

Text generation: Single pause consumes approximately **75 J** of power

Image generation: Single pause consumes approximately **112.5 J** of power

Video generation: Single pause consumes approximately **150 J** of power

When fabricated from shape memory alloy to create warning flowers, they require only 25 J of energy to fully open at room temperature.

This means that with just **one pause**, the wasted power is enough to **fully activate** the warning flower. Frequent pauses will result in excessive energy waste and unnecessary resource consumption.

Temperature elevation induces recovery of memory metals